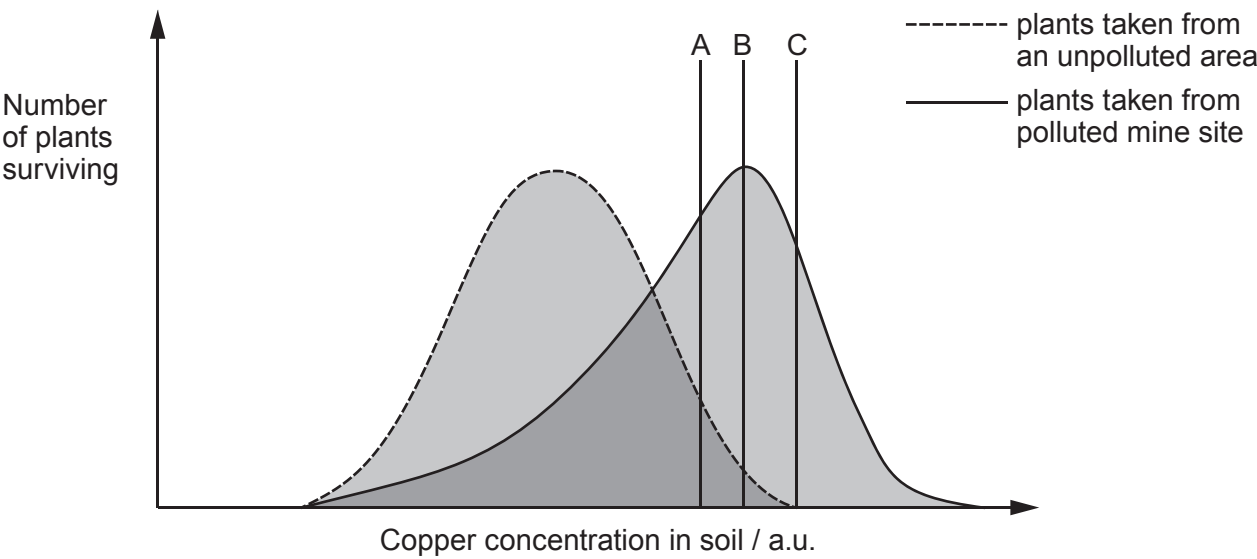


2. Plants, such as sweet vernal grass (*Anthoxanthum odoratum*), require a range of mineral ions which are absorbed from the soil by the roots. Some ions are required in large amounts and others, such as copper, are required in smaller amounts but are toxic at high levels.

One population of sweet vernal grass was found growing in the soil of a derelict mine site in North Wales where copper concentrations reach highly toxic levels. An investigation was carried out to compare these with sweet vernal grass plants from an unpolluted area. Plants from both populations were then grown in soils containing different concentrations of copper. The graph below shows the number of plants from each population that survived at each concentration.



- (a) (i) State the type of variation shown by the plants taken from the unpolluted soil. Explain your answer. [1]

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- (ii) With reference to the graph, state which letter (A, B or C) identifies the mode for sweet vernal grass taken from the mine site and state the difference between the terms *mean* value and *modal* value. [3]

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(b) One hundred years ago the sweet vernal grass was unable to survive on the mine sites but now colonises them. Use your understanding of natural selection to explain this observation. [4]

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(c) Sweet vernal grass plants which are tolerant to high copper concentration in the soil flower at a different time of year from the non-tolerant plants in adjacent areas. Explain why it is predicted that this could lead to the formation of two different species. [4]

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- (d) (i) Another species of grass, common bent (*Agrostis tenuis*), is also copper tolerant and is found growing in polluted mine sites together with sweet vernal grass. The frequency of common bent was seen to be higher than sweet vernal grass. State the type of competition involved and **two** factors for which the plants compete. [2]

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- (ii) State and explain whether high copper concentrations act in a density dependent or density independent way. [1]

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